

# CM0832 - MT5001 Elements of Set Theory

## List of Axioms

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# Outline

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- Preliminaries
- Classification
- Axioms Stating the Existence of Sets
- Axioms Determining Properties of Sets
- Axioms for Building Sets From Other Sets
- References

# Preliminaries

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## Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

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# Classification

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- Axioms stating the existence of sets
- Axioms determining properties of sets
- Axioms for building sets from other sets

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# Axioms Stating the Existence of Sets

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**The Axiom of Existence.** There exists a set which has no elements.

$$(\exists B)(\forall x)(x \notin B).$$

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# Axioms Determining Properties of Sets

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**The Axiom of Extensionality.** If every element of  $X$  is an element of  $Y$  and every element of  $Y$  is an element of  $X$ , then  $X \doteq Y$ .

$$(\forall X)(\forall Y)[(\forall z)(z \in X \leftrightarrow z \in Y) \rightarrow X \doteq Y].$$

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# Axioms for Building Sets From Other Sets

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**The Axiom Schema of Comprehension.** Let  $\mathbf{P}(x)$  be a unary property of  $x$ . For any set  $A$ , there is a set  $B$  such that  $x \in B$  if and only if  $x \in A$  and  $\mathbf{P}(x)$ .

$$(\forall A)(\exists B)(\forall x)(x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x)).$$

**The Axiom of Pair.** For any  $A$  and  $B$ , there is a set  $C$  such that  $x \in C$  if and only if  $x \doteq A$  or  $x \doteq B$ .

$$(\forall A)(\forall B)(\exists C)(\forall x)(x \in C \leftrightarrow x \doteq A \vee x \doteq B).$$

**The Axiom of Union.** For any set  $S$ , there exists a set  $U$  such that  $x \in U$  if and only if  $x \in A$  for some  $A \in S$ .

$$(\forall S)(\exists U)(\forall x)[x \in U \leftrightarrow (\exists A)(x \in A \wedge A \in S)].$$

# Axioms for Building Sets From Other Sets

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**The Axiom of Power Set.** For any set  $S$ , there exists a set  $P$  such that  $X \in P$  if and only if  $X \subseteq S$ .

$$(\forall S)(\exists P)(\forall X)[X \in P \leftrightarrow X \subseteq S].$$

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Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 3).